CAPSTONE PROJECT APPLICATION FORM

(For CMU-SE 450/451 – Software Engineering Program)

1. General information

Item	**Details**
Project Title	AI Voice Assistant Platform for Intelligent IoT Devices
Project Type	☐ Other Industry-Based ☐ Research-Based ☐ Startups/Products ☐: ___
Graduation Course	☑ CMU-SE 450 ☐ CMU-SE 451
Semester/Year	Semester 1/Year 2026-2027
Recommended by	☐ Student Team ☐ , ☑ Lecturers, ☐ Industry Partners

2. Team information

NO.	**Full Name**	**Student ID**	**Email**	**Role (Leadership/Development/ QA/BA/...)**
1	Thinh, Do Ngoc	29211152892	tungdongocdailoc@gmail.com	*Team Member*
2	Tri, Nguyen Huynh	29211155138	tringuyen03112005@gmail.com	*Team Member*
3	Quang, Pham Dinh	29211330920	pdinhquang05@gmail.com	*Team Member*
4	Long, Pham Hoang	29210254319	long290120005@gmail.com	Scrum Master
5	Hoang, Tran Le	29211158794	tranlehoang205@gmail.com	Team leader

3. Supervisor/Instructor

Roles:	**Name**	**Tổ chức**	**Email**
Instructor Instructor	MSc Hoang, Nguyen Thai	**Duy Tan University – International School**	hoang.nguyenthai@axonactive.com
Industry Advisor (if applicable)			

4. Problem Statement

Current AI assistants provide strong conversational capabilities but are commonly accessed through smartphones, computers, or closed commercial ecosystems. This limits the ability to build a customizable physical AI assistant that can be independently managed and extended.

For a Xiaozhi ESP32-based voice assistant, customizing firmware alone is not sufficient for a scalable system. As the number of users and devices increases, the system requires centralized account and device management, authentication, RBAC, device pairing, API Key and quota management, telemetry monitoring, and OTA firmware updates.

The system must also support low-latency voice interaction. Audio from the device needs to be transmitted to a Realtime AI Server, processed through an ASR/STT → LLM → TTS pipeline, and returned to the device as a voice response.

Therefore, the project addresses the need for an integrated AI voice assistant platform that combines customized IoT hardware, realtime AI communication, centralized web-based management, and scalable device management.

Target users / stakeholders:

- General users wishing to interact with AI via voice.
- Users needing to perform quick tasks without using a screen.
- Students/developers interested in researching and developing voice-interactive AI systems.
- System development and administration teams.

- System administrators.

5. Project objectives

The main objective of this project is to design and implement a prototype AI Voice Assistant Platform for Intelligent IoT Devices.

The project objectives are:

- Develop a Web Admin and User Portal for user authentication, RBAC, account management, device management, and authorized configuration.
- Develop a Realtime AI Server that maintains two-way communication with Xiaozhi ESP32 devices and coordinates the ASR/STT → LLM → TTS voice processing pipeline.
- Customize the Xiaozhi ESP32 firmware to support secure server communication, microphone and speaker interaction, device status handling, and OTA update capability.
- Implement device pairing, authentication, online/offline monitoring, heartbeat/telemetry, API Key management, and usage quota control.
- Integrate Cloud AI services in Version 1, including OpenAI API for LLM processing and Cloud ASR/STT and TTS services.
- Design the architecture to support future migration toward self-hosted Local AI models for ASR, LLM, and TTS in Version 2.

6. Proposed solutions

- System Overview: The project builds **the AI Voice Assistant Platform for Intelligent IoT Devices**, a central platform that enables IoT devices capable of receiving and playing audio to connect to AI services through AI Servers.

  Xiaozhi/ESP32 is used as a reference IoT device for deployment and validation of the platform, rather than the entire product of the project.

  Operation Flow:

  User → IoT Device → AI Server → STT/ASR → LLM → TTS → AI Server → IoT Device → User

  In which:

  IoT Device:
  The ESP32/Xiaozhi device records, plays audio, and communicates with the AI Server.

AI Server:
The central component is responsible for device authentication, session management, data reception, and coordination of AI services.

STT/ASR:
Transcribe the user's voice to text.

LLM/AI: Analyze
requirements and generate appropriate responses.

TTS:
Convert text-to-speech responses.

Web Admin:
Allows administrators to manage users, API Keys, devices, and system information.

External APIs:
External services such as Weather, Calendar, Reminder, or others can be integrated.

- Key features of Voice Input:

  - Speech-to-Text.
  - AI Conversational Processing.
  - Text-to-Speech.
  - Voice Response.
  - Conversation Management.
  - User Authentication.
  - API Key Management.
  - IoT Device Management.
  - Device–AI Real-time Communication.
  - Admin Web Dashboard.
  - External API Integration. Expected results:

- Expected results:

  A **complete prototype of the voice AI assistant on Xiaozhi devices**, capable of receiving voice requests, AI processing, and voice responses in real time.

7. Explain complex technical problems

(REQUIRED – for approval)

The project must meet at least three of the following criteria:

Criteria	**Yes/No**	**Explanation**
Conflicting requirements (e.g., security vs. performance)	Yes	The system must strike a balance between AI accuracy, response time, hardware resources, and operating costs.
There is no clear solution available	Yes	Combining Xiaozhi equipment with STT/LLM/TTS components and optimizing the entire pipeline into a unified system requires individual design and testing.
Multi-stakeholder	Yes	Users, development teams, AI/API providers, hardware platforms, and external services.
Multi-technology integration (AI, IoT, Big Data, etc.)	Yes	Combining Embedded/IoT + AI + Speech Processing + Cloud/API + Networking.
Significant social/industrial impact	Yes	Towards a more natural method of human-machine interaction and the ability to be applied in smart assistant devices.

8. System Binding & Impact Analysis

(*CRITICISM SECTION – strong weight)*

8.1 Technical Binding

- ☐ Performance
- ☐ Security & Privacy
- ☐ Scalability
- ☐ Cost
- ☐ Maintainability
- ☐ Others: Network reliability & AI response latency

Explain how the system handles them:

Performance: Optimize the voice processing pipeline and reduce data transfer time between devices, servers and AI services.

Security & Privacy: Limits the data collected, protects API credentials, and controls conversation data.

Scalability: Design the system according to independent modules to be able to replace or extend STT, LLM, TTS and external APIs.

Cost: Prioritize using services and models that fit the prototype's scale, and control the number of requests to AI services.

Maintainability: Separate firmware, AI processing, APIs, and backend components for ease of maintenance.

Network reliability: Handles the case of a loss of connection or an unstable connection between the device and the AI service.

8.2 Global & Social Impact

- Cultural/Social Impact: The system provides a more natural way of interacting with computers through voice and can be extended to multiple languages.
- Environment (green IT): Using a compact hardware appliance and focusing on software/cloud processing can limit the large computing hardware requirements at the device.
- Economic impact: The solution can be developed into an AI assistant platform for IoT devices or smart products with relatively low hardware costs.

8.3 Ethical considerations

- Data privacy: Voice and conversation content may contain personal information, so unnecessary data storage and transmission should be limited.
- AI bias/fairness: Speech recognition results and AI responses can be affected by the user's language, local accent, or pronunciation.

- Intellectual Property: Must comply with the licenses of the firmware, libraries, AI models, and third-party services/APIs used.

9. Engineering & Architectural Stack

Categories	**Recommended technology**
User Interface	Next.js
Backend	Python (FastAPI)
Database	MySQL
DevOps / CI-CD	GitHub + GitHub Actions
Tools (Jira, Git, etc.)	Git, GitHub, VS Code/Antigravity, Jira

10. Development Methods

- ☐ Agile/Scrum
- ☐ V-Model
- ☐ Hybrid

Sprint Plan (Senior):

Sprint 1: System Analysis, Architecture & AI ServerGather and analyze requirements, create SRS, design system architecture and database, develop the Python FastAPI backend, implement user authentication, API Key management, and connect the backend with Microsoft SQL Server.

Sprint 2: IoT & AI Voice IntegrationIntegrate ESP32/Xiaozhi as the reference IoT device, implement WebSocket communication between the IoT device and AI Server, integrate Speech-to-Text (STT/ASR), Large Language Model (LLM), and Text-to-Speech (TTS), and develop the core voice interaction process.

Sprint 3: Admin Web, Integration & TestingDevelop the Next.js Admin Web, implement user and device management, integrate all system components, perform functional and performance testing, fix identified issues, and prepare the final product for demonstration.

11. Main Product Plan

Stage	Product Deliveries
Recommendations	Project Proposals
Design	SRS+SAD
Implementation	Prototype + Testing
Final	Report + Demo

12. Risk Assessment

Risks	Influence	Minimize the impact

13. Expected Contribution

- Technical Contributions
- Innovation/novelty
- Practical application

14. Approval section (for departments)

Role	Name	Decision	Signature	Date
Instructor **Instructor**		☐ Approved ☐ Modifications ☐ Disclaimer		
Program Chair		☐ Approved ☐ Modifications ☐ Disclaimer		

CAPSTONE PROJECT PROPOSAL EVALUATION TABLE

(SE Program – Approval Phase)

Project Name: .....

TeamID: .....

Date: .....

- **Judging Criteria**

Nope.	Criteria	Weight	Primary focus	Score/Score
1	Problem definition	15%	Clear, in line with reality, stakeholders	
2	Complex technical issues	20%	Meets ≥3 complexity characteristics	
3	Feasibility of the solution	15%	Original technical and architectural solidity	
4	Design Constraints	15%	Security, performance, cost, scalability...	
5	Global & Ethical Impact	10%	Ethical, social, environmental factors	
6	Innovation	10%	Novelty, uniqueness	
7	Scope & Planning	10%	Feasible within 14–16 weeks	
8	Team Structure	5%	Roles, balancing competencies	
9	**Overall Score**	**100%**		

Scale: 1 (Poor) – 5 (Excellent)

- **Final Decision**

Overall Score	Decision
≥ 75	✅ Approved
60–74	⚠️ Review & Resubmit
< 60	❌ Disclaimer

- **Comments**

  - Strengths:
  - Weaknesses:
  - Required modifications:

Evaluator's signature